

Heat Kernel in 1d

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Solution of pure initial value problem:

$$u_t = u_{xx}, \quad x \in \mathbb{R}, \quad t > 0$$

$$u(x, 0) = g(x), \quad x \in \mathbb{R}.$$

Fourier Transform basics: For $\xi \in \mathbb{R}$ &

$$g \in C_0^\infty(\mathbb{R}), \quad \hat{g}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{-i\xi x} dx.$$

$$\frac{d\hat{g}}{d\xi}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) (-ix) e^{-i\xi x} dx$$

$$= [(-ix)g]^\wedge(\xi).$$

$$\frac{d^k \hat{g}}{d\xi^k}(\xi) = [(-ix)^k g]^\wedge(\xi).$$

$$\boxed{D^k \mathcal{F}g = \mathcal{F}[(-ix)^k g].}$$

$$\left(\frac{d^k g}{dx^k}\right)^\wedge(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{d^k g}{dx^k}(x) \cdot e^{-i\xi x} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left\{ e^{-i\xi x} \cdot \frac{d^{k-1} g}{dx^{k-1}}(x) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} (-i\xi) e^{-i\xi x} \frac{d^{k-1} g}{dx^{k-1}}(x) dx \right\}$$

$$= (i\xi)^k \hat{g}(\xi). \quad (2)$$

$$\mathcal{F} D^k g = (i\xi)^k \mathcal{F} g.$$

Schwartz class: (Class of rapidly decaying fns.)
 $\mathcal{S} \equiv \{ u \in C^\infty(\mathbb{R}) : |x|^k |D^\alpha u(x)| \text{ is bounded on } \mathbb{R} \}$
 $\forall k \in \mathbb{N} \text{ and } \alpha \in \mathbb{N} \}$.

$g \in \mathcal{S} \Rightarrow \hat{g}$ exists because g is rapidly decaying fn at ∞ .

Lemma: ① If $g \in L^1(\mathbb{R})$ then $\hat{g}$ is bounded.

② If $g \in \mathcal{S}$ then $\hat{g} \in \mathcal{S}$.

Proof: ① $|\hat{g}(\xi)| \leq \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} |g(x)| dx = M < \infty$.

$$\textcircled{2} \quad \frac{d\hat{g}}{d\xi}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{-ix\xi} (-ix) dx$$

Because $g \in \mathcal{S}$, $|xg(x)|$ is bounded and we have $-ixg(x)$ rapidly decays at ∞ .

So $\hat{g}$ is differentiable.

In fact $\hat{g} \in C^\infty$ and $\frac{d^k \hat{g}}{d\xi^k}(\xi) = [(-ix)^k g]^\wedge(\xi)$.

Similarly for $g \in \mathcal{S}$ we have

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$$\left(\frac{d^k g}{dx^k}\right)^\wedge(\xi) = (i\xi)^k \hat{g}(\xi).$$

Now to show $\hat{g} \in \mathcal{S}$ we have to show that $|\xi|^k |D_\xi^\alpha \hat{g}(\xi)|$ is bounded on $\mathbb{R}$.

$$\begin{aligned} |\xi^k D_\xi^\alpha \hat{g}(\xi)| &= |\xi^k [(-ix)^\alpha g]^\wedge(\xi)| \\ &= \left| \xi^k \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (-ix)^\alpha g(x) e^{-i\xi x} dx \right| \end{aligned}$$

$$\left| [D_x^k x^\alpha g]^\wedge(\xi) \right| = \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} D_x^k x^\alpha g(x) \cdot e^{-i\xi x} dx \right|$$

$$\begin{aligned} \text{By integration by parts} \quad &= \left| \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^\alpha g(x) (i\xi)^k e^{-i\xi x} dx \right| \\ &= \left| \xi^k \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (-ix)^\alpha g(x) e^{-i\xi x} dx \right| \end{aligned}$$

$$\therefore |\xi^k D_\xi^\alpha \hat{g}(\xi)| = |[D_x^k x^\alpha g]^\wedge(\xi)|.$$

$$D_x^k x^\alpha g \in L^1(\mathbb{R}) \text{ why?}$$

$g \in C^\infty \Rightarrow x^\alpha g \in C^\infty \Rightarrow$ All derivatives of $x^\alpha g$ exist and are continuous.

$\Rightarrow D_x^k x^\alpha g$ is integrable. ④

By part ① of Lemma $(D_x^k x^\alpha g)^\wedge(\xi)$ is bounded. So $\xi^k D_\xi^\alpha \hat{g}(\xi)$ is bounded

Hence $\hat{g} \in \mathcal{S}$.

Define inverse Fourier transform for $g \in L^1(\mathbb{R})$:

$$g^\vee(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(\xi) e^{ix\xi} d\xi, \quad x \in \mathbb{R}.$$

Like the Lemma, we can prove that

$$\hat{g} \in \mathcal{S} \Rightarrow (\hat{g})^\vee \in \mathcal{S}.$$

Fourier Inversion Theorem: If $g \in \mathcal{S}$ then $(\hat{g})^\vee = g$.

$$\begin{aligned} g(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{g}(\xi) e^{ix\xi} d\xi \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(y) \cdot e^{-iy\xi} dy \right] e^{ix\xi} d\xi \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(y) \cdot e^{i(x-y)\xi} dy d\xi. \end{aligned}$$

The above theorem is true for more functions than in $\mathcal{S}$, but not true in general.

Solution of the pure IVP:

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$$u_t = u_{xx}, \quad x \in \mathbb{R}, t > 0.$$

$$u(x, 0) = g(x), \quad x \in \mathbb{R}.$$

Assume $g(x) \in C_0^\infty(\mathbb{R})$.

Taking Fourier transform w.r.t. x variable to the heat equation we get.

$$\frac{\partial \hat{u}}{\partial t}(\xi, t) = -\xi^2 \hat{u}(\xi, t).$$

$$\Rightarrow \hat{u}(\xi, t) = C e^{-\xi^2 t}.$$

$$\Rightarrow \hat{u}(\xi, 0) = \hat{g}(\xi)$$

$$\hat{u}(\xi, t) = \hat{g}(\xi) e^{-\xi^2 t}.$$

Taking inverse Fourier Transform.

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{g}(\xi) e^{-\xi^2 t} e^{ix\xi} d\xi.$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} e^{i(x-y)\xi - \xi^2 t} d\xi \right] g(y) dy$$

$$K(x, y, t) = \frac{1}{\sqrt{4\pi t}} e^{-\frac{(x-y)^2}{4t}}.$$

This is called heat kernel or Gaussian.